

Bayesian Multivariate Linear Models

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Multivariate Linear Regression

- Multivariate linear regression generalizes the standard linear model to predict *multiple* correlated outcomes simultaneously.
- It addresses the following question: How does a set of quantities of primary interest, $y = (y_1, y_2, \dots, y_q)$, vary jointly as a function of explanatory variables x ?
- In general, we are interested in the conditional distribution of a q -dimensional outcome vector y , given x , parametrized as $p(y \mid \theta, x)$.
- Using a multivariate approach allows us to borrow strength across different outcomes by explicitly modeling their covariance structure.

Data Setup

- We have a set of experimental subjects $i = 1, 2, \dots, n$.
- For each unit, we measure a q -dimensional outcome vector $y_i^\top = (y_{i1}, \dots, y_{iq})$ and p explanatory variables $x_i^\top = (1, x_{i1}, \dots, x_{i,p-1})$.
- We collect the outcomes into an $n \times q$ matrix Y and the explanatory variables into an $n \times p$ matrix X :

$$Y = \begin{pmatrix} y_1^\top \\ y_2^\top \\ \vdots \\ y_n^\top \end{pmatrix}; \quad X = \begin{pmatrix} x_1^\top \\ x_2^\top \\ \vdots \\ x_n^\top \end{pmatrix}.$$

- The regression coefficients are now a $p \times q$ matrix β , and the error variance is captured by a $q \times q$ covariance matrix Σ .

Matrix Normal and Inverse-Wishart Distributions

- A random $p \times q$ matrix β follows a Matrix Normal distribution, denoted $\beta \sim MN_{p,q}(\mu, \Phi, \Sigma)$, if:

$$p(\beta) \propto |\Sigma|^{-p/2} |\Phi|^{-q/2} \exp \left\{ -\frac{1}{2} \text{tr} [\Sigma^{-1}(\beta - \mu)^\top \Phi^{-1}(\beta - \mu)] \right\}$$

where μ is the $p \times q$ mean matrix, Φ is a $p \times p$ row covariance, and Σ is a $q \times q$ column covariance.

- A $q \times q$ positive definite matrix Σ follows an Inverse-Wishart distribution, denoted $\Sigma \sim IW(\nu, \Psi)$, if:

$$p(\Sigma) \propto |\Sigma|^{-(\nu+q+1)/2} \exp \left\{ -\frac{1}{2} \text{tr} (\Sigma^{-1} \Psi) \right\}$$

where $\nu > q - 1$ is the degrees of freedom and Ψ is the $q \times q$ scale matrix.

Conjugate Bayesian Multivariate Linear Regression

- A conjugate Bayesian multivariate linear model (the MNIW model) is given by:

$$Y \mid \beta, \Sigma, X \sim MN_{n,q}(X\beta, I_n, \Sigma)$$

$$\beta \mid \Sigma \sim MN_{p,q}(M_0 m_0, M_0, \Sigma)$$

$$\Sigma \sim IW(\nu_0, \Psi_0)$$

- Unknown parameters include the regression coefficient matrix and the covariance matrix, i.e. $\theta = \{\beta, \Sigma\}$.
- The likelihood of the data is:

$$p(Y \mid \beta, \Sigma) \propto |\Sigma|^{-n/2} \exp \left\{ -\frac{1}{2} \text{tr} [\Sigma^{-1} (Y - X\beta)^T (Y - X\beta)] \right\}$$

- The joint posterior density is

$$p(\beta, \Sigma \mid Y) \propto p(\Sigma) \times p(\beta \mid \Sigma) \times p(Y \mid \beta, \Sigma).$$

Deriving the Joint Posterior $p(\beta, \Sigma | Y)$

$$\begin{aligned} p(\beta, \Sigma | Y) &\propto IW(\Sigma | \nu_0, \Psi_0) \times MN(\beta | M_0 m_0, M_0, \Sigma) \times MN(Y | X\beta, I_n, \Sigma) \\ &\propto |\Sigma|^{-\frac{\nu_0+q+1}{2}} \exp \left\{ -\frac{1}{2} \text{tr} (\Sigma^{-1} \Psi_0) \right\} \\ &\quad \times |\Sigma|^{-\frac{p}{2}} \exp \left\{ -\frac{1}{2} \text{tr} [\Sigma^{-1} (\beta - M_0 m_0)^\top M_0^{-1} (\beta - M_0 m_0)] \right\} \\ &\quad \times |\Sigma|^{-\frac{n}{2}} \exp \left\{ -\frac{1}{2} \text{tr} [\Sigma^{-1} (Y - X\beta)^\top (Y - X\beta)] \right\} \\ &\propto |\Sigma|^{-\frac{\nu_0+n+p+q+1}{2}} \exp \left\{ -\frac{1}{2} \text{tr} (\Sigma^{-1} [\Psi_0 + Q(\beta, Y)]) \right\} \end{aligned} \quad (1)$$

where the quadratic terms in β are combined as:

$$Q(\beta, Y) = (\beta - M_0 m_0)^\top M_0^{-1} (\beta - M_0 m_0) + (Y - X\beta)^\top (Y - X\beta) .$$

Matrix Completion of Square

Let us expand and recombine $Q(\beta, Y)$, noting that M_0 is symmetric, so $(M_0 m_0)^\top M_0^{-1} \beta = m_0^\top M_0 M_0^{-1} \beta = m_0^\top \beta$:

$$\begin{aligned} Q(\beta, Y) &= (\beta - M_0 m_0)^\top M_0^{-1} (\beta - M_0 m_0) + (Y - X\beta)^\top (Y - X\beta) \\ &= \beta^\top M_0^{-1} \beta - \beta^\top m_0 - m_0^\top \beta + m_0^\top M_0 m_0 \\ &\quad + Y^\top Y - Y^\top X\beta - \beta^\top X^\top Y + \beta^\top X^\top X\beta \\ &= \beta^\top (M_0^{-1} + X^\top X) \beta - \beta^\top (m_0 + X^\top Y) \\ &\quad - (m_0^\top + Y^\top X) \beta + (m_0^\top M_0 m_0 + Y^\top Y) \\ &= (\beta - Mm)^\top M^{-1} (\beta - Mm) + S^* \end{aligned} \tag{2}$$

where we define the updated parameters M , m , and S^* as:

$$\begin{aligned} M^{-1} &= M_0^{-1} + X^\top X \\ m &= m_0 + X^\top Y \\ S^* &= m_0^\top M_0 m_0 + Y^\top Y - m^\top Mm \end{aligned}$$

Note that M , m , and S^* do not depend upon β .

Matrix Completion of Square (Details)

- Let us revisit how the quadratic in the previous slide simplifies. We grouped the expanded trace into three blocks:

$$\text{Quadratic in } \beta : \quad \beta^\top (M_0^{-1} + X^\top X) \beta$$

$$\text{Linear in } \beta : \quad -\beta^\top (m_0 + X^\top Y) - (m_0^\top + Y^\top X) \beta$$

$$\text{Constant (w.r.t } \beta) : \quad m_0^\top M_0 m_0 + Y^\top Y$$

- Substituting $M^{-1} = M_0^{-1} + X^\top X$ and $m = m_0 + X^\top Y$ we get:

$$\beta^\top M^{-1} \beta - \beta^\top m - m^\top \beta + m_0^\top M_0 m_0 + Y^\top Y$$

- We complete the perfect matrix square $(\beta - Mm)^\top M^{-1}(\beta - Mm)$ by adding and subtracting the cross-term $(Mm)^\top M^{-1}Mm = m^\top Mm$:

$$\begin{aligned} & \left[\beta^\top M^{-1} \beta - \beta^\top m - m^\top \beta + m^\top Mm \right] - m^\top Mm + m_0^\top M_0 m_0 + Y^\top Y \\ &= (\beta - Mm)^\top M^{-1}(\beta - Mm) + \underbrace{m_0^\top M_0 m_0 + Y^\top Y - m^\top Mm}_{S^*} \end{aligned}$$

Factorizing the Joint Posterior

- The posterior is proportional to the joint: $p(\beta, \Sigma | Y) \propto p(\beta, \Sigma, Y)$.
- Substitute (2) back into the joint posterior (1):

$$\begin{aligned} p(\beta, \Sigma | Y) &\propto |\Sigma|^{-\frac{\nu_0 + n + p + q + 1}{2}} \exp \left\{ -\frac{1}{2} \text{tr} \left(\Sigma^{-1} \left[\Psi_0 + S^* + (\beta - Mm)^\top M^{-1} (\beta - Mm) \right] \right) \right\} \\ &\propto \left(|\Sigma|^{-\frac{\nu + q + 1}{2}} \exp \left\{ -\frac{1}{2} \text{tr} \left[\Sigma^{-1} \Psi \right] \right\} \right) \\ &\quad \times \left(|\Sigma|^{-\frac{p}{2}} \exp \left\{ -\frac{1}{2} \text{tr} \left[\Sigma^{-1} (\beta - Mm)^\top M^{-1} (\beta - Mm) \right] \right\} \right) \end{aligned}$$

where we define the posterior Inverse-Wishart parameters:

$$\nu = \nu_0 + n \quad \text{and} \quad \Psi = \Psi_0 + S^* = \Psi_0 + m_0^\top M_0 m_0 + Y^\top Y - m^\top M m.$$

- This directly factors into $IW(\Sigma | \nu, \Psi) \times MN(\beta | Mm, M, \Sigma)$.

Exact Posterior Sampling

- The joint posterior density factorizes perfectly:

$$p(\beta, \Sigma | Y) \propto \underbrace{IW(\Sigma | \nu, \Psi)}_{p(\Sigma | Y)} \times \underbrace{MN(\beta | Mm, M, \Sigma)}_{p(\beta | \Sigma, Y)},$$

where

$$\nu = \nu_0 + n; \quad \Psi = \Psi_0 + Y^\top Y + m_0^\top M_0 m_0 - m^\top M m;$$

$$m = m_0 + X^\top Y; \quad M^{-1} = M_0^{-1} + X^\top X.$$

Algorithm Exact posterior sampling from $p(\beta, \Sigma | Y)$

- 1: **for** (i in $1 : N$), **do**
 - 2: Draw $\Sigma_{(i)} \sim IW(\nu, \Psi)$
 - 3: Draw $\beta_{(i)} \sim MN(Mm, M, \Sigma_{(i)})$
 - 4: **end for**
 - 5: **return** $(\Sigma_{(i)}, \beta_{(i)})_{i=1,2,\dots,N}$
-

Bayesian Predictions from Multivariate Regression

- Let $\tilde{Y}$ denote the $m \times q$ matrix of predictive outcomes corresponding to predictors $\tilde{X}$ ($m \times p$).
- Joint posterior simplifies due to $\tilde{Y} \perp Y \mid \{\beta, \Sigma\}$:

$$p(\beta, \Sigma, \tilde{Y} \mid Y) \propto \underbrace{p(\beta, \Sigma \mid Y)}_{MNIW(\beta, \Sigma \mid Mm, M, \nu, \Psi)} \times \underbrace{p(\tilde{Y} \mid \beta, \Sigma, Y)}_{p(\tilde{Y} \mid \beta, \Sigma) = MN(\tilde{Y} \mid \tilde{X}\beta, I_m, \Sigma)}$$

- We draw $\tilde{Y}$ from the likelihood for every posterior sample:

Algorithm Exact posterior predictive sampling from $p(\tilde{Y} \mid Y)$

```
1: for ( $i$  in  $1 : N$ ), do
2:   Draw  $\Sigma_{(i)} \sim IW(\nu, \Psi)$ 
3:   Draw  $\beta_{(i)} \sim MN(Mm, M, \Sigma_{(i)})$ 
4:   Draw  $\tilde{Y}_{(i)} \sim MN(\tilde{X}\beta_{(i)}, I_m, \Sigma_{(i)})$ 
5: end for
6: return  $(\tilde{Y}_{(i)})_{i=1,2,\dots,N}$ 
```

Bayesian Regression Analysis for BIG DATA

Bayesian updating with independent data

- Sequential update for $\theta = \{\beta, \Sigma\}$: Update posterior for $t = 1, 2, \dots, T$

$$\begin{aligned} p(\theta \mid Y_1, \dots, Y_t) &\propto p(\theta, Y_1, \dots, Y_{t-1}, Y_t) \\ &\propto \underbrace{p(\theta, Y_1, \dots, Y_{t-1})}_{p(Y_1, \dots, Y_{t-1}) \times p(\theta \mid Y_1, \dots, Y_{t-1})} \times \underbrace{p(Y_t \mid \theta, Y_1, \dots, Y_{t-1})}_{p(Y_t \mid \theta)} \\ &\propto p(\theta \mid Y_1, \dots, Y_{t-1}) \times p(Y_t \mid \theta) . \end{aligned}$$

- Note the use of independent data given θ :

$$p(Y_t \mid \theta, Y_1, \dots, Y_{t-1}) = p(Y_t \mid \theta)$$

- Posterior at time $t - 1$ becomes the prior for the next t .
- Divide & Conquer by loading manageable blocks of data for each t .

Bayesian regression using Divide and Conquer

- Partition data as $\{Y_t, X_t\}$, $t = 1, 2, \dots, T$, where Y_t is $n_t \times q$, X_t is $n_t \times p$ and $n = \sum_{t=1}^T n_t$.
- Suppose we assume block independence (with optional known row-covariance V_t):

$$Y_t | \beta, \Sigma \stackrel{ind}{\sim} MN_{n_t, q}(X_t \beta, V_t, \Sigma);$$
$$\beta | \Sigma \sim MN_{p, q}(M_0 m_0, M_0, \Sigma); \quad \Sigma \sim IW(\nu_0, \Psi_0) .$$

- Let $\theta = \{\beta, \Sigma\}$. The posterior is $p(\theta | Y) \propto p(\theta, Y)$:

$$IW(\Sigma | \nu_0, \Psi_0) \times MN(\beta | M_0 m_0, M_0, \Sigma) \times \prod_{t=1}^T MN(Y_t | X_t \beta, V_t, \Sigma) .$$

- Note: If each data point is independent, then $V_t = I_{n_t}$. Otherwise, we can design data collection such that data are independent across blocks.

Bayesian regression using Divide and Conquer

- Write the posterior distribution at time $t - 1$ as

$$p(\theta \mid Y_1, \dots, Y_{t-1}) = IW(\Sigma \mid \nu_{t-1}, \Psi_{t-1}) \times MN(\beta \mid M_{t-1}m_{t-1}, M_{t-1}, \Sigma)$$

- Posterior at time t updates as

$$p(\theta \mid Y_1, \dots, Y_t) = IW(\Sigma \mid \nu_t, \Psi_t) \times MN(\beta \mid M_t m_t, M_t, \Sigma)$$

where ν_t , Ψ_t , m_t and M_t are updated as follows:

$$m_t = m_{t-1} + X_t^\top V_t^{-1} Y_t; \quad M_t^{-1} = M_{t-1}^{-1} + X_t^\top V_t^{-1} X_t$$

$$\nu_t = \nu_{t-1} + n_t; \quad \Psi_t = \Psi_{t-1} + Q_t, \quad \text{where}$$

$$Q_t = Y_t^\top V_t^{-1} Y_t + m_{t-1}^\top M_{t-1} m_{t-1} - m_t^\top M_t m_t.$$

Bayesian analysis for BIG DATA (algorithm)

Algorithm Exact posterior sampling for Divide & Conquer

```
1: Set  $m \leftarrow m_0$ ,  $M^{-1} \leftarrow M_0^{-1}$ ,  $\nu \leftarrow \nu_0$ ,  $\Psi \leftarrow \Psi_0$ 
2:  $M \leftarrow \text{solve}(M^{-1}, I_p)$ 
3: for ( $t$  in  $1 : T$ ), do
4:    $Q \leftarrow Y_t^\top V_t^{-1} Y_t + m^\top M m$ 
5:    $m \leftarrow m + X_t^\top V_t^{-1} Y_t$  and  $M^{-1} \leftarrow M^{-1} + X_t^\top V_t^{-1} X_t$ 
6:    $M \leftarrow \text{solve}(M^{-1}, I_p)$ 
7:    $Q \leftarrow Q - m^\top M m$ 
8:    $\Psi \leftarrow \Psi + Q$  and  $\nu \leftarrow \nu + n_t$ 
9: end for
10:
11: for ( $i$  in  $1 : N$ ), do
12:   Draw  $\Sigma_{(i)} \sim IW(\nu, \Psi)$ 
13:   Draw  $\beta_{(i)} \sim MN(Mm, M, \Sigma_{(i)})$ 
14: end for
15: return  $\{\beta_{(i)}, \Sigma_{(i)}\}_{i=1, \dots, N}$ 
```


Bayesian analysis for BIG DATA (algorithm using linear algebra)

Algorithm Exact posterior sampling for Divide & Conquer

```
1: Set  $m \leftarrow m_0$ ,  $M^{-1} \leftarrow M_0^{-1}$ ,  $\nu \leftarrow \nu_0$ ,  $\Psi \leftarrow \Psi_0$ 
2:  $L^{-1} \leftarrow \text{cholesky}(M^{-1})$  and  $L = \text{trsolve}(L^{-1}, I_p)$ 
3: for ( $t$  in  $1 : T$ ), do
4:    $Q \leftarrow Y_t^\top V_t^{-1} Y_t + (Lm)^\top (Lm)$ 
5:    $m \leftarrow m + X_t^\top V_t^{-1} Y_t$  and  $M^{-1} \leftarrow M^{-1} + X_t^\top V_t^{-1} X_t$ 
6:    $L^{-1} \leftarrow \text{cholesky}(M^{-1})$  and  $L = \text{trsolve}(L^{-1}, I_p)$ 
7:    $Q \leftarrow Q - (Lm)^\top (Lm)$ 
8:    $\Psi \leftarrow \Psi + Q$  and  $\nu \leftarrow \nu + n_t$ 
9: end for
10: for ( $i$  in  $1 : N$ ), do
11:   Draw  $\Sigma_{(i)} \sim IW(\nu, \Psi)$  and  $Z \sim MN_{p,q}(0, I_p, I_q)$ 
12:    $\Sigma_{(i)}^{1/2} \leftarrow \text{cholesky}(\Sigma_{(i)})$ 
13:    $\beta_{(i)} \leftarrow L^\top Lm + L^\top Z(\Sigma_{(i)}^{1/2})^\top$ 
14: end for
15: return  $\{\beta_{(i)}, \Sigma_{(i)}\}_{i=1, \dots, N}$ 
```

Federated Learning: Decentralized Divide & Conquer

- Federated Learning paradigm: A central server communicates with T clients (local servers). The raw data (Y_t, X_t) cannot leave client t .
- The central server holds prior parameters: M_0^{-1} , m_0 , ν_0 , and Ψ_0 .
- To achieve the exact same global posterior without centralizing the data, we exploit the additive nature of Bayesian linear regression using conjugate priors.
- Instead of passing a sequentially updated prior from one block to the next, the server broadcasts the global prior, clients compute local summary statistics in parallel, and the server aggregates them in one shot.

Federated Learning: Local Client Computations

- The client computes three localized sufficient statistics.
- Precision Matrix Update: $\Delta M_t^{-1} = X_t^\top V_t^{-1} X_t$.
- Mean Matrix Update: $\Delta m_t = X_t^\top V_t^{-1} Y_t$.
- Sum of Squares Matrix: $\Delta Q_t = Y_t^\top V_t^{-1} Y_t$.
- Client t then sends only the tuple $(\Delta M_t^{-1}, \Delta m_t, \Delta Q_t, n_t)$ back to the central server.
- This preserves privacy and vastly reduces communication overhead. For a dataset with millions of rows but $p = 10$ predictors and $q = 5$ outcomes, the client only transmits a 10×10 matrix, a 10×5 matrix, a 5×5 matrix, and one scalar.

Federated Learning: Server Aggregation

- The global precision and natural mean matrices are:

$$M_T^{-1} = M_0^{-1} + \sum_{t=1}^T \Delta M_t^{-1} \quad m_T = m_0 + \sum_{t=1}^T \Delta m_t .$$

- Degrees of freedom for the Inverse-Wishart distribution updates:

$$\nu_T = \nu_0 + \sum_{t=1}^T n_t$$

- Sum across all T clients *telescopes* to yield Ψ_T :

$$\sum_{t=1}^T Q_t = \left(\sum_{t=1}^T \Delta Q_t \right) + m_0^\top M_0 m_0 - m_T^\top M_T m_T ; \quad \Psi_T = \Psi_0 + \sum_{t=1}^T Q_t .$$

- The server computes the global scale matrix in one step:

$$\Psi_T = \Psi_0 + \sum_{t=1}^T \Delta Q_t + m_0^\top M_0 m_0 - m_T^\top M_T m_T .$$

Global Posterior Sampling

- With M_T , m_T , ν_T , and Ψ_T fully assembled at the central server, the global posterior is mathematically indistinguishable from a model that centralized all data in one place.

Algorithm Draw N samples from $p(\beta, \Sigma \mid Y)$

```
1: for ( $i$  in  $1 : N$ ), do  
2:   Draw  $\Sigma_{(i)} \sim IW(\nu_T, \Psi_T)$   
3:   Draw  $\beta_{(i)} \sim MN(M_T m_T, M_T, \Sigma_{(i)})$   
4: end for  
5: return  $(\Sigma_{(i)}, \beta_{(i)})_{i=1,2,\dots,N}$ 
```
